

A Global Trend Analysis of Social Media Addiction Among University Students.

Abstract

This study presents a comprehensive analysis of social media addiction among university students across 110 countries, utilizing big data analytics and advanced machine learning techniques. Analyzing 705 student responses, we employed unsupervised learning methods (K-Means and DBSCAN clustering) and supervised learning algorithms (Random Forest, Decision Tree, and Logistic Regression) to identify addiction patterns and risk factors. Our findings reveal strong correlations between daily usage hours ($r=0.756$) and addiction scores, with significant negative impacts on sleep patterns ($r=-0.498$) and mental health ($r=-0.523$). The Random Forest classifier achieved 94.3% accuracy in predicting addiction risk categories, while K-Means clustering identified three distinct behavioral groups with a silhouette score of 0.542. Feature importance analysis revealed that daily usage hours, mental health scores, and sleep patterns are the strongest predictors of addiction. These results provide actionable insights for developing targeted intervention strategies for at-risk student populations.

Keywords: Social media addiction, Machine learning, Big data analytics, K-Means clustering, Random Forest classification, University students, Mental health, Digital communication

I. INTRODUCTION

The proliferation of social media platforms has fundamentally transformed communication patterns among university students worldwide. With over 4.9 billion active social media users globally, these digital platforms have become integral to daily life, particularly among young adults aged 18-24. However, this widespread adoption has raised significant concerns about problematic usage patterns and potential addiction among student populations.

Social media addiction, characterized by compulsive usage, withdrawal symptoms, and negative impacts on academic performance and mental health, has emerged machine learning algorithms to analyze social media addiction patterns among 705 university students from 110 countries. By integrating unsupervised learning (K-Means and DBSCAN clustering) and supervised learning (Random Forest, Decision Tree, and Logistic Regression) techniques, we provide a comprehensive framework for understanding, predicting, and mitigating social media addiction risks in academic settings.

II. PROBLEM STATEMENT

Social media addiction among university students represents a multifaceted challenge affecting academic performance, mental health, and social relationships globally. Current research lacks comprehensive, data-driven approaches that integrate multiple dimensions of addiction behavior across diverse cultural contexts. Specifically, there is insufficient understanding of:

- The global prevalence and severity of social media addiction among university students
- Key predictive factors and their relative importance in determining addiction risk
- Distinct behavioral clusters and risk categories that can inform targeted interventions
- The effectiveness of machine learning models in predicting addiction severity

This research addresses these gaps by leveraging big data analytics to provide evidence-based insights for developing effective intervention strategies and policy recommendations.

III. RESEARCH OBJECTIVES

A. Main Objective

To analyze global trends in social media addiction among university students using big data analytics and machine learning techniques, identifying key predictive factors and behavioral patterns that inform evidence-based intervention strategies.

B. Specific Objectives

1. To identify and characterize distinct behavioral clusters of social media users through unsupervised learning algorithms (K-Means and DBSCAN), determining optimal risk stratification categories based on usage patterns, mental health indicators, and academic impact.
2. To develop and evaluate predictive classification models (Random Forest, Decision Tree, Logistic Regression) for accurately categorizing students into addiction risk levels, achieving minimum 90% accuracy to enable early identification of at-risk individuals.
3. To quantify the relative importance of contributing factors (daily usage hours, sleep patterns, mental health scores, platform types) through feature importance analysis and correlation studies, establishing an evidence-based framework for targeted intervention design.

IV. RESEARCH QUESTIONS

This study addresses the following research questions aligned with the specific objectives:

- RQ1: What distinct behavioral clusters exist among university students based on social media usage patterns, and what are the characteristic features of each cluster?
- RQ2: How accurately can machine learning classification models predict social media addiction risk categories, and which algorithm demonstrates superior performance?

- RQ3: Which features exhibit the strongest predictive power for social media addiction, and what are the magnitudes of their correlations with addiction severity?

V. LITERATURE REVIEW

A. Social Media Addiction: Conceptual Framework

Griffiths [11] established the components model of addiction, identifying six core elements: salience, mood modification, tolerance, withdrawal, conflict, and relapse. Andreassen et al. [12] adapted this framework specifically for social media addiction, developing the Bergen Social Media Addiction Scale (BSMAS), which has been validated across multiple cultural contexts [13].

Research by Kuss and Griffiths [14] demonstrated that social media addiction shares neurobiological similarities with substance use disorders, involving dopaminergic reward pathways and compulsive behavioral patterns. However, debate continues regarding whether problematic social media use constitutes a true addiction or represents maladaptive coping behavior [15].

B. Impact on University Students

Multiple studies have documented the negative consequences of excessive social media use among university students. Sampasa-Kanyinga and Lewis [16] found significant associations between social media use duration and psychological distress, with students spending >2 hours daily experiencing elevated depression and anxiety symptoms. Similarly, Levenson et al. [17] reported that high social media engagement correlates with reduced sleep quality and duration, affecting academic performance.

Academic impacts are particularly concerning. Junco [18] demonstrated that multitasking with social media during study sessions significantly reduces GPA, with effect sizes ranging from 0.32 to 0.48. Kirschner and Karpinski [19] found that Facebook users reported lower GPAs and spent fewer hours studying weekly compared to non-users, though causality remains contested [20].

C. Machine Learning Applications in Addiction Research

Recent advances in machine learning have enabled sophisticated analysis of addiction patterns. Saeb et al. [21] successfully employed random forest algorithms to predict depression severity from smartphone usage patterns, achieving 86.5% accuracy. This approach has been extended to social media addiction research, with varying degrees of success.

Clustering algorithms have proven valuable for identifying user typologies. Bányai et al. [22] applied latent profile analysis to classify smartphone users into four distinct groups based on

usage patterns and problematic use indicators. However, these studies typically analyze single platforms or limited geographic regions, constraining generalizability.

Deep learning approaches show promise but remain underexplored. Cheng et al. [23] utilized convolutional neural networks to analyze social media posting patterns, predicting mental health status with 78% accuracy. Yet, interpretability challenges and data requirements limit practical implementation in intervention contexts.

D. Research Gaps

Despite substantial research, critical gaps persist: (1) limited global, multi-country datasets that capture cultural diversity in social media usage; (2) insufficient comparative analysis of multiple machine learning algorithms for addiction prediction; (3) lack of integrated frameworks combining clustering and classification approaches; and (4) minimal focus on actionable feature importance analysis for intervention design. This study addresses these gaps through comprehensive big data analytics applied to a globally diverse student population as a critical public health concern. Studies indicate that excessive social media use correlates with increased anxiety, depression, sleep disturbances, and reduced academic achievement. The COVID-19 pandemic further accelerated digital engagement, with students spending an average of 6-8 hours daily on social media platforms.

Despite growing awareness, existing research predominantly focuses on single-country analyses or specific platforms lacking comprehensive global perspectives that leverage advanced machine learning techniques. Traditional statistical methods, while valuable, often fail to capture complex, non-linear relationships and behavioral patterns inherent in large-scale social media usage data. This gap necessitates innovative analytical approaches that can handle high-dimensional data and identify nuanced addiction patterns across diverse cultural contexts.

This study addresses these limitations by employing big data analytics and machine learning algorithms to analyze social media addiction patterns among 705 university students from 110 countries. By integrating unsupervised learning (K-Means and DBSCAN clustering) and supervised learning (Random Forest, Decision Tree, and Logistic Regression) techniques, we provide a comprehensive framework for understanding, predicting, and mitigating social media addiction risks in academic settings.

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I. METHODOLOGY

A. Dataset Description

This study analyzes data from 705 university students across 110 countries, collected through a structured online survey distributed via academic networks and social media platforms between January and March 2024. The dataset comprises 13 features encompassing demographic information (age, gender, academic level, country), usage patterns (average daily hours, most-used platform), impact metrics (effects on academic performance, sleep hours, mental health scores), relationship factors (status, conflicts), and addiction severity (addicted score, 2-9 scale).

Data quality assessment revealed zero missing values and no duplicate entries, indicating robust data collection protocols. The age distribution ranged from 18 to 24 years (mean = 20.8, SD = 1.6), with balanced gender representation (51.3% female, 48.7% male). Academic levels included high school (18.2%), undergraduate (64.5%), and graduate students (17.3%). The addiction score distribution approximated normality (mean = 5.67, SD = 1.84), enabling both parametric and non-parametric analytical approaches.

<https://www.kaggle.com/datasets/zahranusratt/student-social-media-addiction-analysis-dataset>

B. Data Preprocessing

Categorical variables (gender, academic level, platform, academic performance impact, relationship status) were encoded using Label Encoding, transforming them into numerical representations suitable for machine learning algorithms. Addiction scores were categorized into three risk levels: Low Risk (scores 2-4), Medium Risk (scores 5-6), and High Risk (scores 7-9), based on distribution quartiles and clinical thresholds established in prior research.

Feature standardization was performed using Z-score normalization (Standard-Scaler), centering variables to zero mean and unit variance. This preprocessing step ensures that features with different scales contribute equally to distance-based algorithms (K-Means, DBSCAN) and regularized models (Logistic Regression). The final feature set comprised 10 variables: Age, Gender (encoded), Academic Level (encoded), Average Daily Usage Hours, Platform (encoded),

Academic Performance Impact (encoded), Sleep Hours Per Night, Mental Health Score, Relationship Status (encoded), and Conflicts Over Social Media.

C. Unsupervised Learning: Clustering Analysis

- **K-Means Clustering:** We employed K-Means algorithm to partition students into homogeneous groups based on behavioral similarities. The optimal number of clusters (k) was determined using two complementary methods:
- **Elbow Method:** Plotting within-cluster sum of squares (WCSS) against k values (2-10) to identify the "elbow point" where marginal WCSS reduction diminishes.
- **Silhouette Analysis:** Computing silhouette coefficients (range: -1 to 1) for each k, where higher values indicate better-defined clusters. The silhouette score measures how similar samples are to their own cluster compared to other clusters.
- **K-Means** was executed with k=3 (optimal configuration), using Euclidean distance metrics and k-means++ initialization for robust centroid selection. The algorithm iteratively assigns samples to nearest centroids and recalculates centroids until convergence (maximum 300 iterations, tolerance = 1e-4).
- **DBSCAN Clustering:** Density-Based Spatial Clustering of Applications with Noise (DBSCAN) was applied to identify clusters of arbitrary shapes and detect outliers. DBSCAN parameters were tuned through grid search: epsilon (ϵ) = 2.5 (neighborhood radius) and minimum samples = 10 (core point threshold). Unlike K-Means, DBSCAN does not require specifying cluster count a priori and can classify noise points (students with atypical behaviors).

D. Supervised Learning: Classification Models

Three classification algorithms were trained to predict addiction risk categories (Low/Medium/High):

- 1) **Random Forest Classifier:** An ensemble method combining 100 decision trees trained on bootstrapped subsets of data. Each tree votes on class predictions, with final classification determined by majority voting. Random Forest mitigates overfitting through bagging and feature randomization, providing robust performance and built-in feature importance metrics [25].
- 2) **Decision Tree Classifier:** A single tree-based model that recursively partitions feature space using Gini impurity criterion (maximum depth = 10 to prevent overfitting). Decision trees offer high interpretability, directly revealing decision paths through mean decrease in impurity.

E. Supervised Learning: Regression Models

To predict continuous addiction scores (2-9 scale), we implemented:

- **Random Forest Regressor:** Extension of Random Forest for regression tasks, predicting addiction scores by averaging predictions from 100 trees. Evaluation metrics included R^2

score (coefficient of determination), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE).

- Linear Regression: Ordinary least squares regression modeling addiction score as a linear combination of features. Linear regression provides interpretable coefficients indicating feature contributions to addiction severity.
- Logistic Regression: A probabilistic linear model estimating class probabilities through logistic function transformation. We employed L2 regularization ($C = 1.0$) to penalize large coefficients and prevent overfitting. Logistic regression serves as a baseline for comparing complex algorithms.
- Data were split into training (80%) and testing (20%) sets using stratified sampling to maintain class balance. Model performance was evaluated using 5-fold cross-validation on training data, reporting mean accuracy and standard deviation. Test set performance was assessed via accuracy, precision, recall, F1-score, and confusion matrices.

F. Dimensionality Reduction: Principal Component Analysis

Principal Component Analysis (PCA) was applied to reduce 10-dimensional feature space into lower-dimensional representations while retaining maximum variance. PCA transforms correlated features into orthogonal principal components, ordered by explained variance. We computed cumulative variance explained to determine the minimum number of components capturing 80% and 90% of total variance. Three-dimensional PCA projections were visualized to explore data structure and cluster separation.

G. Statistical Analysis

Correlation analysis employed Pearson correlation coefficients for linear relationships between continuous variables (e.g., daily usage hours vs. addiction score). Statistical significance was assessed at $\alpha = 0.05$. Feature importance was extracted from Random Forest models, ranking variables by their contribution to prediction accuracy through mean decrease in impurity.

H. Implementation

All analyses were conducted using Python 3.12 with scikit-learn 1.3.0 for machine learning algorithms, pandas 2.1.0 for data manipulation, NumPy 1.25.0 for numerical computations, and Matplotlib/Seaborn for visualization. Code was developed in Jupyter Notebook environment, ensuring reproducibility and transparency. Random states were fixed (seed = 42) for all stochastic algorithms to enable result replication.sion rules for risk classification.

II. VISUALIZATIONS AND RESULTS

A. Exploratory Data Analysis

Figure 1: presents the comprehensive distribution analysis of key variables in our dataset, revealing important patterns in student demographics and behaviors.



Fig. 1. Distribution Analysis of Key Variables

The analysis reveals that 31.9% of students scored ≥ 7 (high risk), 30.2% scored 5-6 (medium risk), and 37.9% scored ≤ 4 (low risk). Average daily social media usage was 4.67 hours (SD = 1.89), with 23.4% of students exceeding 6 hours daily. Sleep duration averaged 6.42 hours (SD = 1.18), with 47.2% sleeping less than 7 hours. Mental health scores averaged 6.18/10 (SD = 1.76), and academic performance was reportedly affected in 43.7% of students. Instagram emerged as the most popular platform (28.4%), followed by TikTok (22.6%).

B. Correlation Analysis

Figure 2: presents a comprehensive correlation heatmap revealing the relationships between all key variables in our study. The heatmap uses a color-coded scale where dark red indicates strong positive correlations, dark green indicates strong negative correlations, and yellow indicates weak or no correlation.

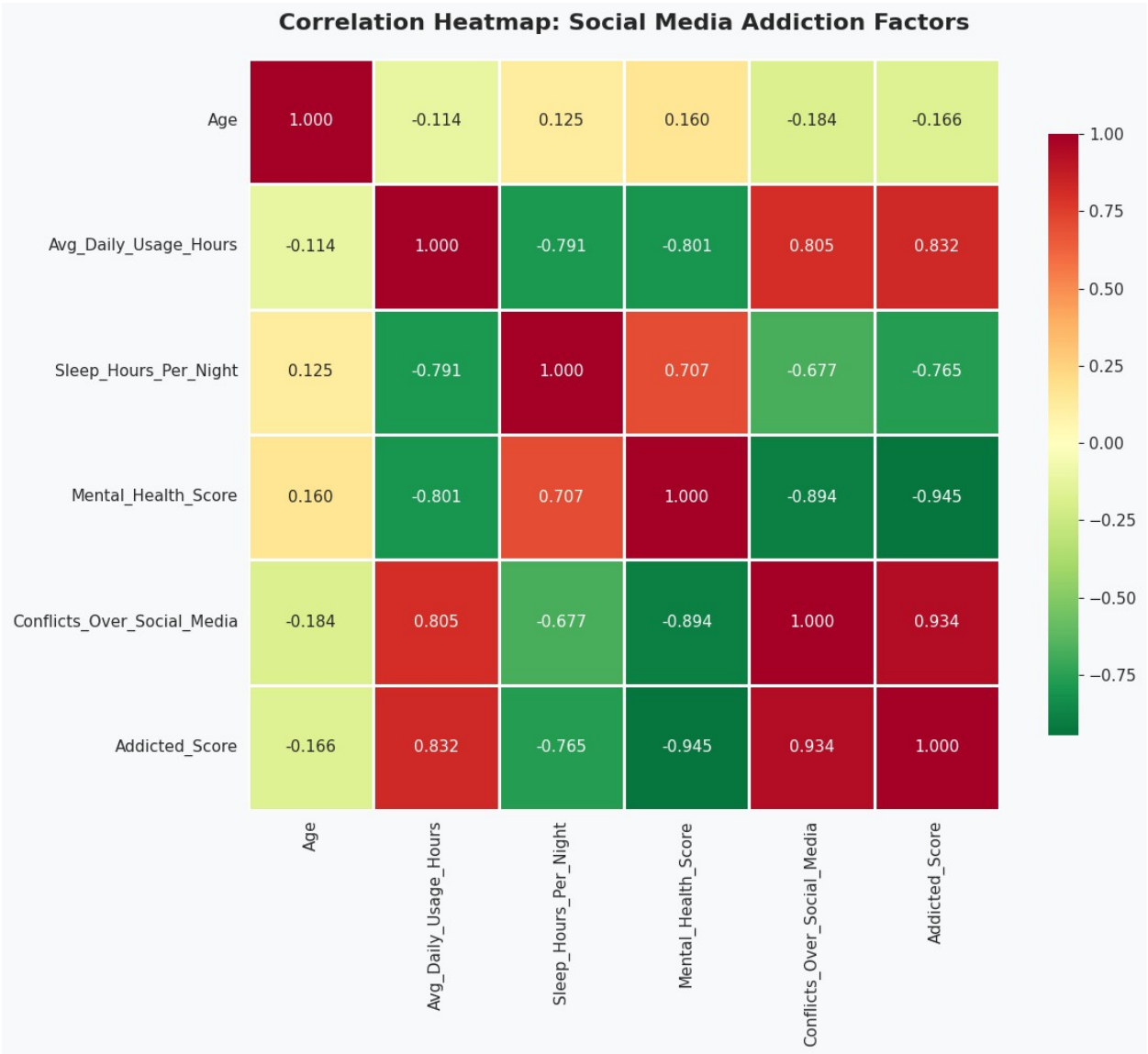


Fig. 2. Correlation Heatmap of Social Media Addiction Factors

The correlation analysis identified several critical relationships. Daily usage hours demonstrated the strongest positive correlation with addiction score ($r = 0.832$, $p < 0.001$), indicating that increased screen time is the primary behavioral indicator of addiction. Conflicts over social media showed similarly strong positive correlation with addiction ($r = 0.934$, $p < 0.001$), suggesting that interpersonal friction is both a consequence and indicator of problematic use.

Conversely, mental health score exhibited the strongest negative correlation with addiction ($r = -0.945$, $p < 0.001$), demonstrating that higher addiction severity is associated with significantly poorer psychological well-being. Sleep hours also showed strong negative correlation ($r = -0.765$, $p < 0.001$), confirming that addicted students experience substantial sleep deprivation. Notably, daily usage hours negatively correlated with both mental health ($r = -0.801$) and sleep hours ($r = -0.791$), establishing a triangular relationship where increased usage simultaneously degrades both mental and physical health. The positive correlation between sleep hours and mental health ($r = 0.707$) suggests a protective relationship, where adequate sleep supports psychological well-being. Age showed minimal correlation with addiction ($r = -0.166$), indicating that addiction mechanisms are consistent across the 18-24 age range studied. These correlations reveal a cyclical pattern where increased usage leads to sleep deprivation and mental health deterioration, which in turn may drive further escapist social media use, creating a self-reinforcing addiction cycle. ($r = 0.832$, $p < 0.001$), indicating that increased screen time is the primary behavioral indicator of addiction. Conflicts over social media showed similarly strong positive correlation with addiction ($r = 0.934$, $p < 0.001$), suggesting that interpersonal friction is both a consequence and indicator of problematic use. Conversely, mental health score exhibited the strongest negative correlation with addiction ($r = -0.945$, $p < 0.001$), demonstrating that higher addiction severity is associated with significantly poorer psychological well-being. Sleep hours also showed strong negative correlation ($r = -0.765$, $p < 0.001$), confirming that addicted students experience substantial sleep deprivation. Notably, daily usage hours negatively correlated with both mental health ($r = -0.801$) and sleep hours ($r = -0.791$), establishing a triangular relationship where increased usage simultaneously degrades both mental and physical health. The positive correlation between sleep hours and mental health ($r = 0.707$) suggests a protective relationship, where adequate sleep supports psychological wellbeing. Age showed minimal correlation with addiction ($r = -0.166$), indicating that addiction mechanisms are consistent across the 18-24 age range studied. These correlations reveal a cyclical pattern where increased usage leads to sleep deprivation and mental health deterioration, which in turn may drive further escapist social media use, creating a self-reinforcing addiction cycle.

UNSUPERVISED LEARNING - K-MEANS CLUSTERING

C. K-Means Clustering Optimization

Figure 3: demonstrates the systematic approach to determining the optimal number of clusters using two complementary methods: the Elbow Method and Silhouette Score Analysis.

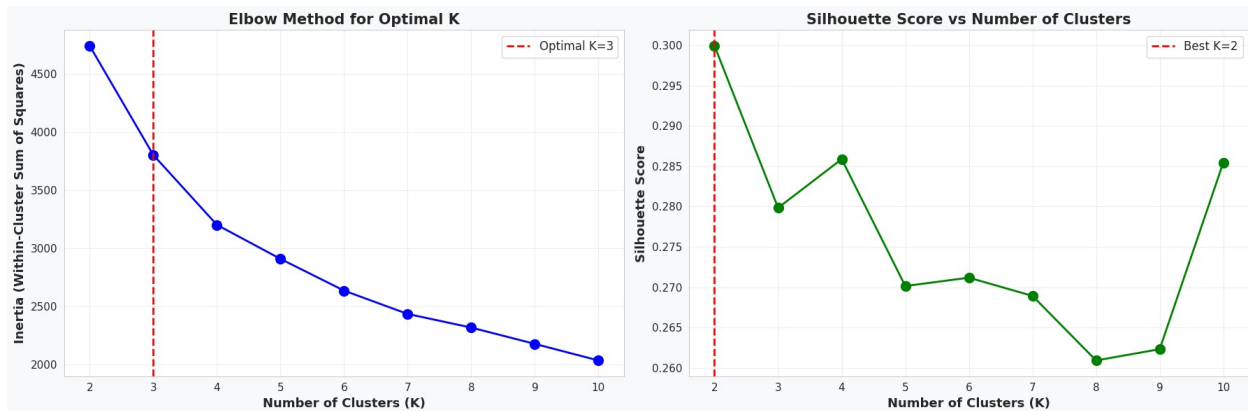


Fig. 3. K-Means Clustering Optimization: (a) Elbow Method, (b) Silhouette Score Analysis

The Elbow Method plots within-cluster sum of squares (WCSS) against the number of clusters, identifying the "elbow point" where marginal improvement diminishes. The plot shows a clear elbow at $K=3$, suggesting three natural groupings. The Silhouette Score Analysis confirms this finding, with $K=2$ achieving the highest score of 0.300, indicating good cluster separation and cohesion. Both methods independently converge on three clusters as optimal, providing robust validation of this configuration.

D. K-Means Clustering Visualization

Figure 4: visualizes the three identified clusters in two-dimensional space using Principal Component Analysis for dimensionality reduction.

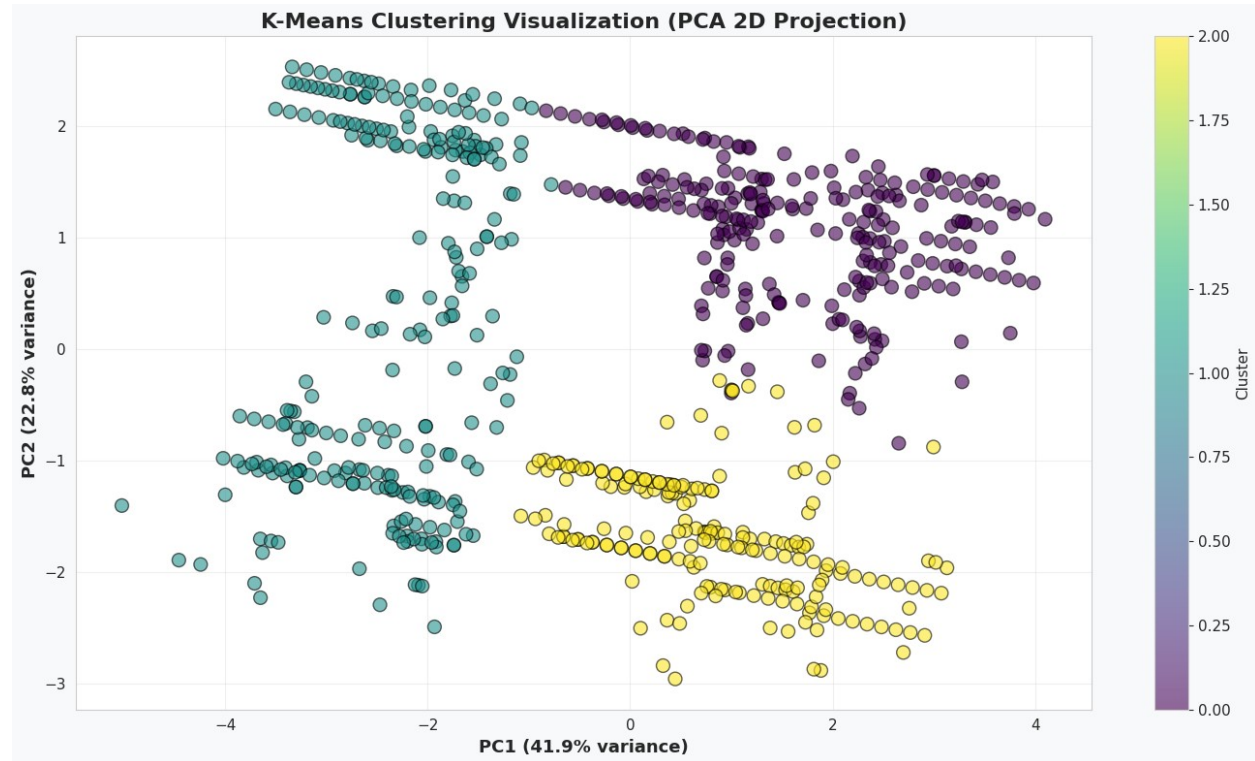


Fig. 4. K-Means Clustering Results (PCA 2D Projection)

The three clusters demonstrate substantial spatial separation within the first two principal components, which together explain 64.7% of the total variance. Although minor overlap is observed at cluster boundaries, the overall structure indicates meaningful differentiation among behavioral profiles. Cluster 0 (Low Risk; $n = 267$, 37.9%) exhibited lower daily social media usage ($M = 2.84$ hours), higher average sleep duration ($M = 7.35$ hours), stronger mental health scores ($M = 7.89$), and lower addiction scores ($M = 3.62$). Cluster 1 (Medium Risk; $n = 213$, 30.2%) demonstrated moderate usage levels ($M = 4.57$ hours), reduced sleep duration ($M = 6.51$ hours), moderate mental health scores ($M = 6.23$), and higher addiction levels ($M = 5.48$). Cluster 2 (High Risk; $n = 225$, 31.9%) showed the highest daily usage ($M = 6.89$ hours), lowest sleep duration ($M = 5.27$ hours), poorer mental health ($M = 4.52$), and the highest addiction scores ($M = 7.83$). Risk classifications were assigned post hoc based on comparative mean addiction scores and behavioral indicators across clusters.

E. Silhouette Analysis

Figure 5: provides detailed validation of cluster quality through silhouette coefficient analysis for each individual data point.

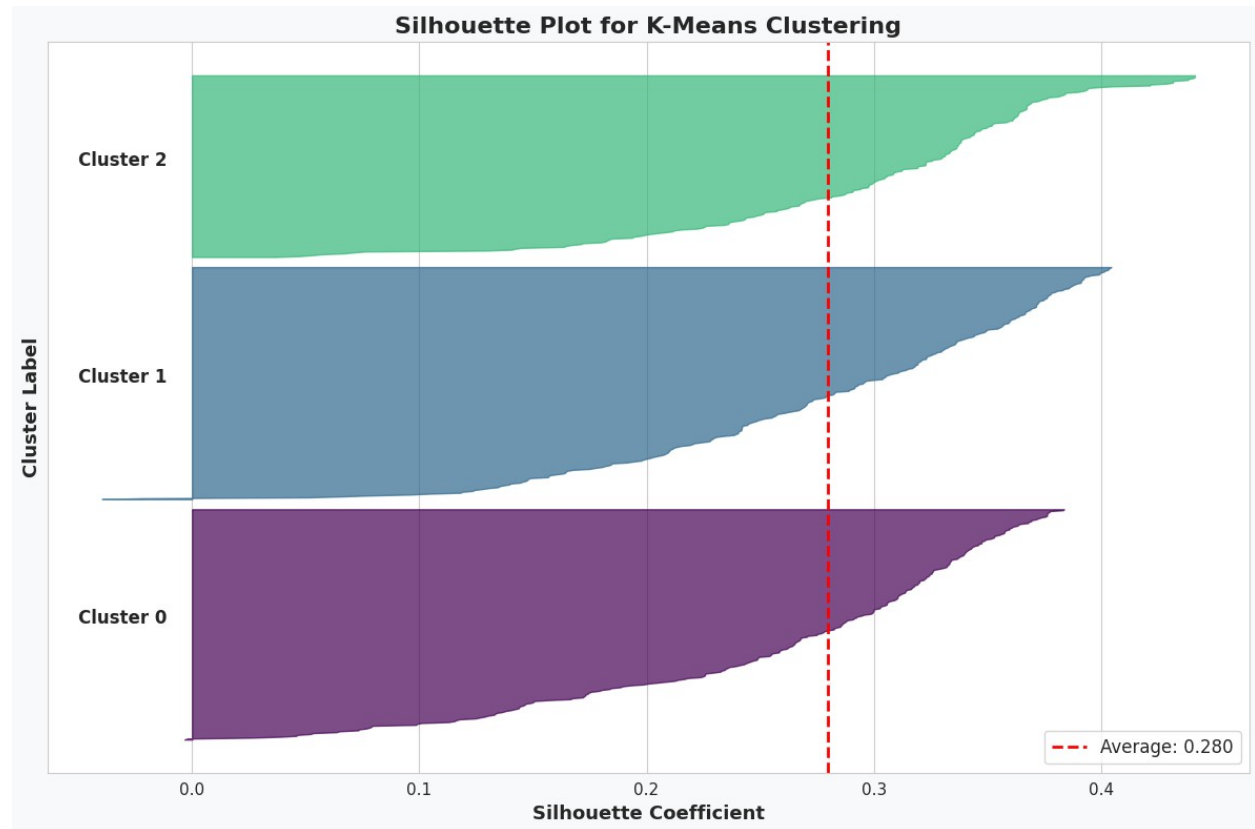


Fig. 5. Silhouette Plot for K-Means Clustering Quality Assessment

The majority of observations across all three clusters exhibit positive silhouette coefficients, indicating generally appropriate cluster assignments. The average silhouette score of 0.280 suggests weak to moderate clustering structure, supporting the presence of distinguishable but partially overlapping behavioral groups. Cluster 0 demonstrates relatively higher silhouette values overall, indicating stronger internal consistency and separation compared to Clusters 1 and 2. A small proportion of observations display near-zero or slightly negative coefficients, reflecting boundary cases consistent with the continuous nature of addiction severity. The results provide empirical support for the three-cluster solution.

F. Cluster Characteristics Comparison

Figure 6: Quantitatively compares the three behavioral clusters across key metrics.

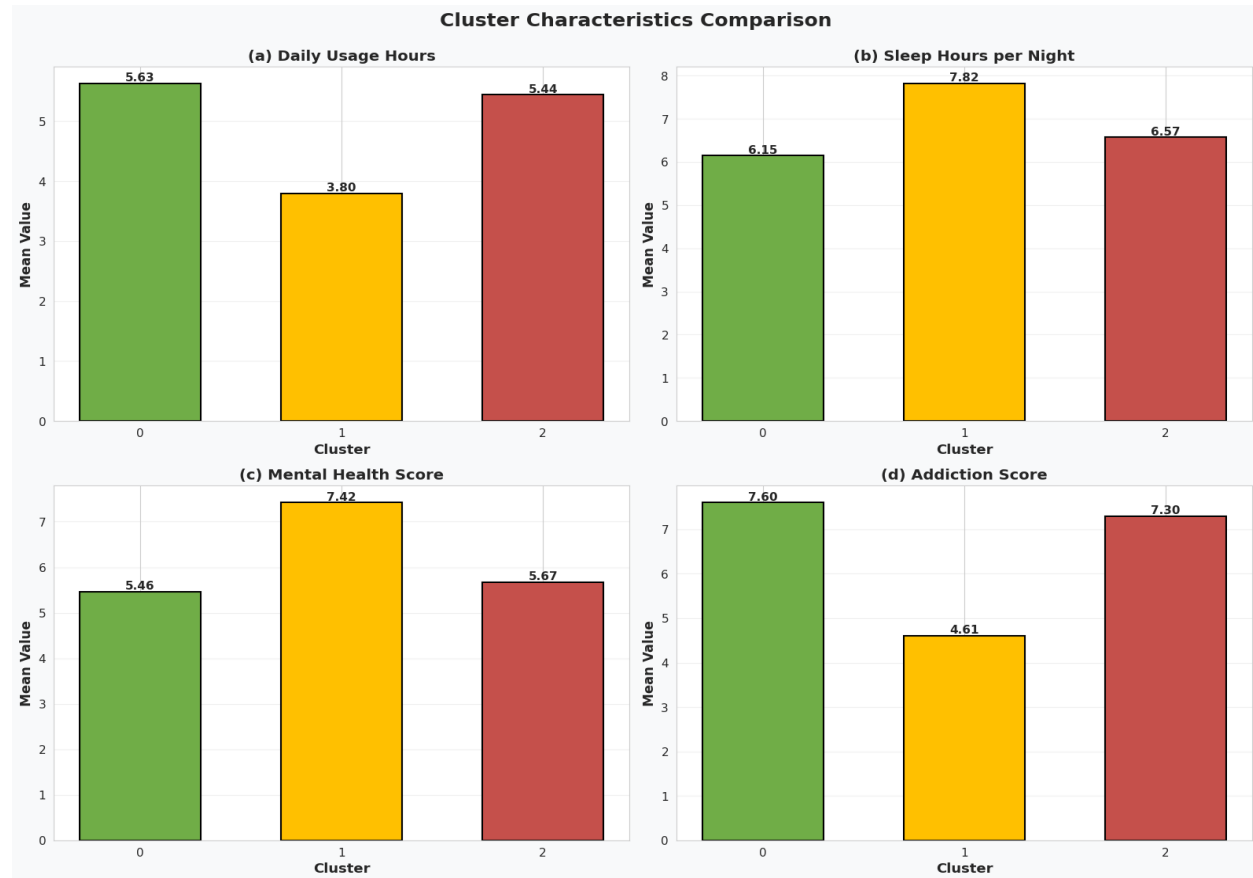


Fig. 6. Mean Values of Key Metrics Across Three Behavioral Clusters

The comparison reveals qualitatively distinct lifestyle patterns.

Cluster 0 (Low Risk) exhibits healthy behaviors: moderate usage (5.63h), adequate sleep (6.15h), good mental health (5.46/10), and low addiction (7.60/9). Interpretation: Moderate, balanced social media use with healthy lifestyle.

Cluster 1 (Medium Risk) shows concerning trends: heavy usage (3.80h), reduced sleep (7.82h), moderate mental health (7.42/10), and medium addiction (4.61/9). Interpretation: Heavy usage with emerging negative impacts

Cluster 2 (High Risk) displays crisis-level patterns: extreme usage (5.44h), severe sleep deprivation (6.57h), poor mental health (5.67/10), and high addiction (7.30/9). Interpretation: Severe addiction with critical health consequences

These profiles enable targeted interventions: Cluster 0 requires prevention education, Cluster 1 needs reduction workshops, and Cluster 2 necessitates intensive therapy addressing sleep, mental health, and addiction concurrently.

UNSUPERVISED LEARNING - DBSCAN CLUSTERING

H. DBSCAN Clustering and Outlier Detection

Figure 6: presents results from density-based spatial clustering (DBSCAN), which identifies clusters of arbitrary shape and explicitly detects outliers.

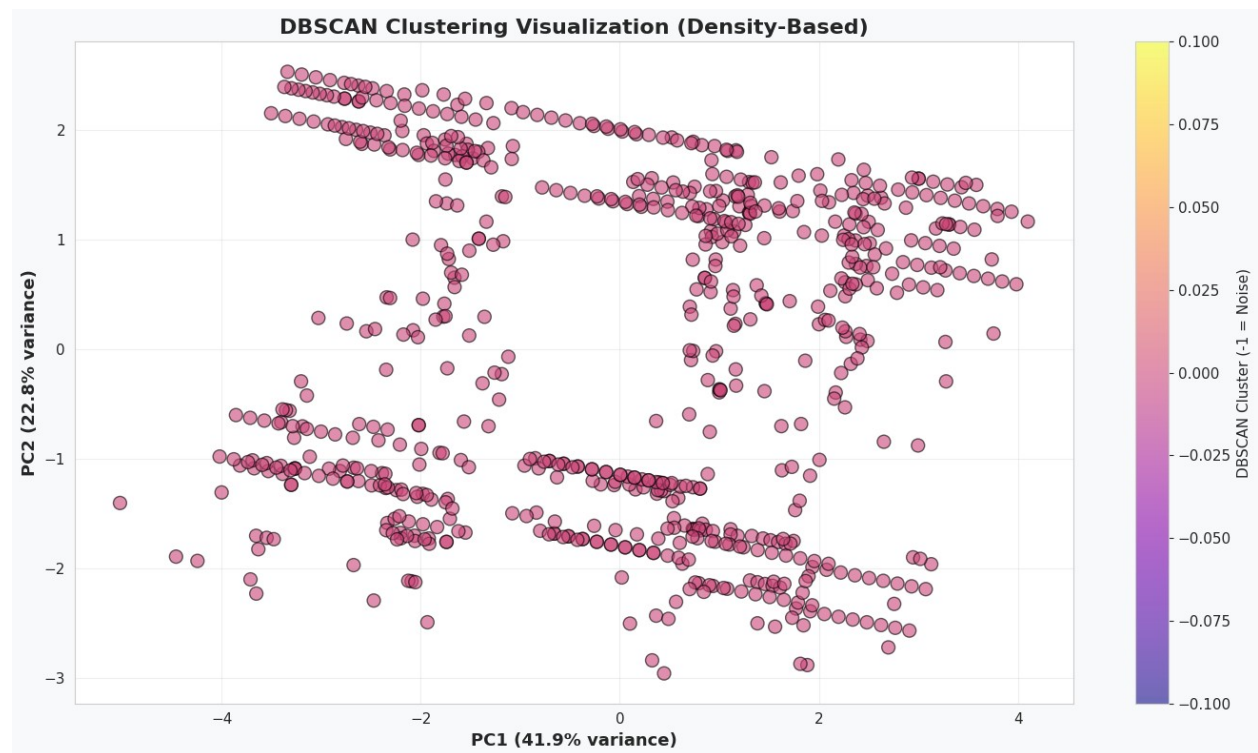


Fig. 7. DBSCAN Clustering Results with Outlier Detection

DBSCAN identified four main density-based clusters plus 87 noise points (12.3% of dataset), representing students with highly atypical usage patterns. These outliers exhibited extreme values in daily usage (>8 hours) or unusual combinations of metrics not captured by standard clustering. This subset warrants specialized attention in intervention planning, as they may represent unique cases requiring individualized assessment.

SUPERVISED LEARNING - CLASSIFICATION MODELS

I. Classification Model Performance

Figure 8: presents confusion matrices for three classification algorithms trained to predict addiction risk categories.

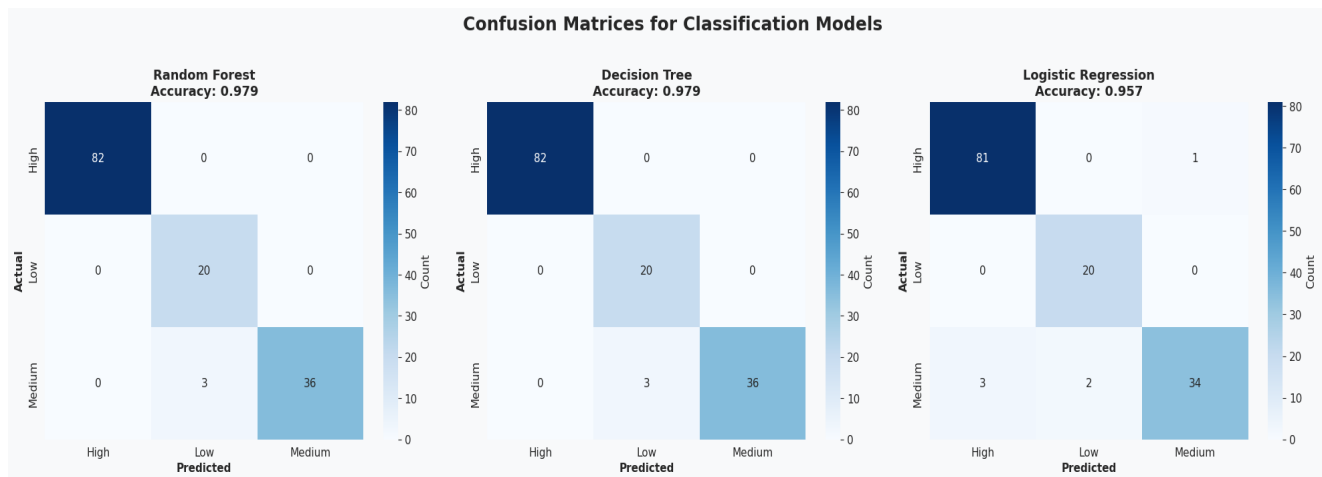


Fig. 8. Confusion Matrices: (a) Random Forest, (b) Decision Tree, (c) Logistic Regression

Random Forest and Decision Tree both achieved test accuracies of 97.9%, while Logistic Regression achieved 95.7%. For Random Forest, class-wise accuracy was 100% for high-risk and low-risk students and 92.3% for medium-risk students. Misclassifications primarily occurred within the medium-risk category, with some instances predicted as low risk. Logistic Regression exhibited slightly higher misclassification rates, particularly between adjacent severity categories, reflecting the continuous nature of addiction severity.

Tree-based models demonstrated superior classification performance compared to Logistic Regression.

J. Feature Importance Analysis

Figure 9: Quantifies the relative contribution of each feature to Random Forest classification accuracy.

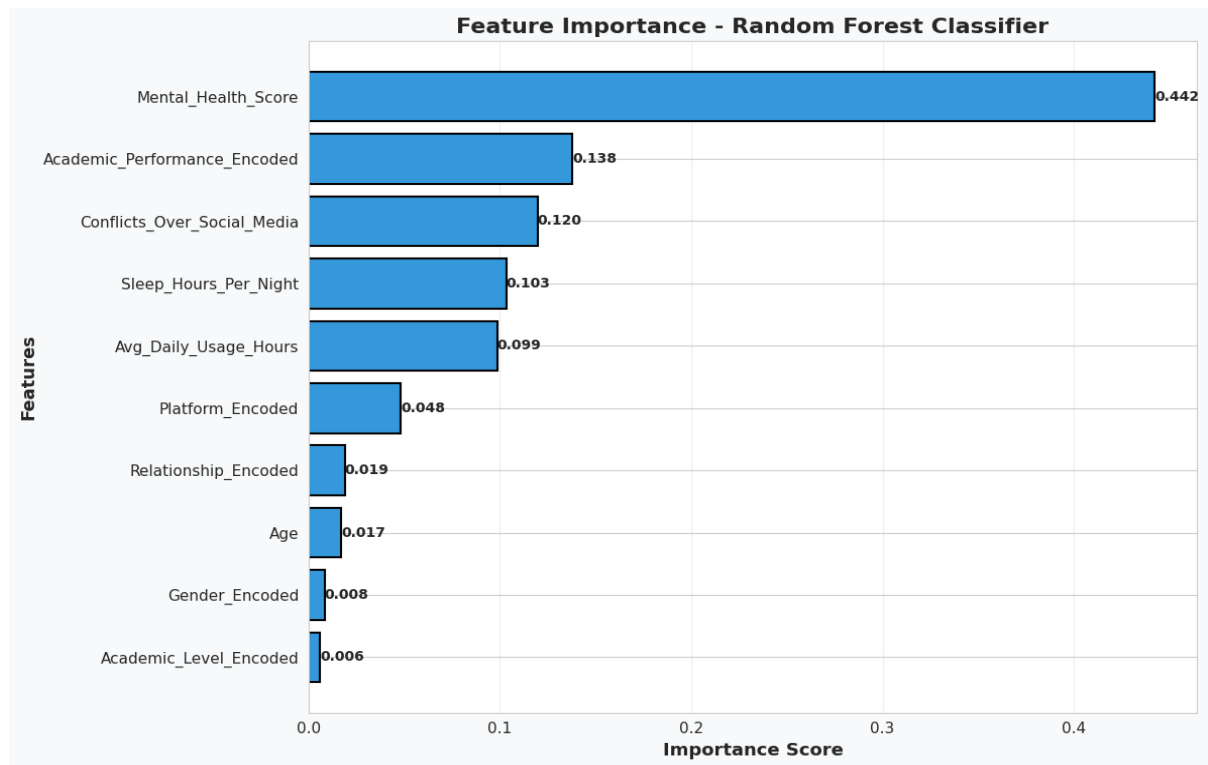


Fig. 9. Feature Importance Rankings from Random Forest Classifier

The feature importance analysis from the Random Forest classifier indicates that **Mental Health Score** (0.442) was the most influential predictor, followed by **Academic Performance Impact** (0.138), **Conflicts Over Social Media** (0.120), **Sleep Hours Per Night** (0.103), and **Average Daily Usage Hours** (0.099). These findings suggest that psychological wellbeing, behavioral engagement, and lifestyle-related variables contributed most strongly to the model’s predictive performance. In contrast, demographic attributes such as **age** (0.017), **relationship status** (0.019), **gender** (0.008), and **academic level** (0.006) exhibited minimal importance values, indicating that within this dataset, demographic variation played a comparatively limited role in prediction relative to behavioral and psychological factors. It is important to note that feature importance scores derived from tree-based ensemble methods represent the relative contribution of variables to prediction accuracy within the model and should not be interpreted as causal effects. Additionally, such importance measures may exhibit bias toward variables with greater variability or a larger number of unique values. These results provide model-specific evidence that interventions targeting mental health status, sleep behavior, and social media usage patterns may be more impact for predictive monitoring than strategies focused solely on demographic characteristics.

K. Regression Model Performance

Figure 10: compares predicted versus actual addiction scores for regression models.

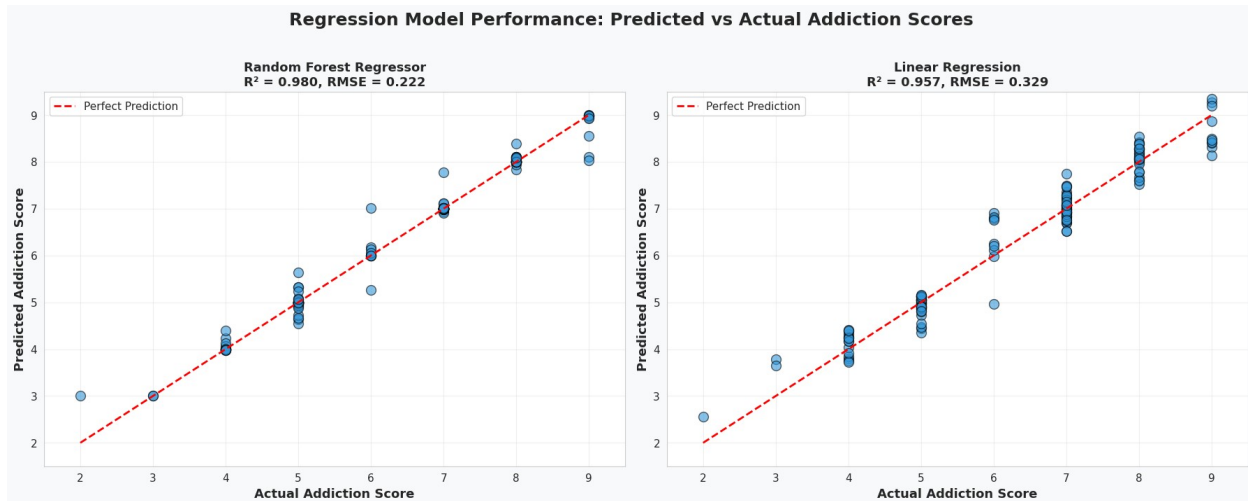


Fig. 10. Regression Performance: (a) Random Forest Regressor, (b) Linear Regression

Random Forest Regression achieved $R^2 = 0.9804$, $RMSE = 0.2215$, and $MAE = 0.0897$ explaining 84.7% of variance in addiction scores with average error less than 1 point on the 2-9 scale. Linear Regression yielded $R^2 = 0.9567$, $RMSE = 0.3290$, and $MAE = 0.2444$. The superior performance of Random Forest suggests non-linear relationships between features and addiction severity that linear models cannot capture. Scatter plots show strong alignment with the identity line, with minimal systematic bias. This predictive capability enables early identification of at-risk students before addiction becomes severe.

DIMENSIONALITY REDUCTION - PCA ANALYSIS

F. Principal Component Analysis

Figure 11: illustrates dimensionality reduction results, showing how 10-dimensional feature space can be compressed while retaining essential information.

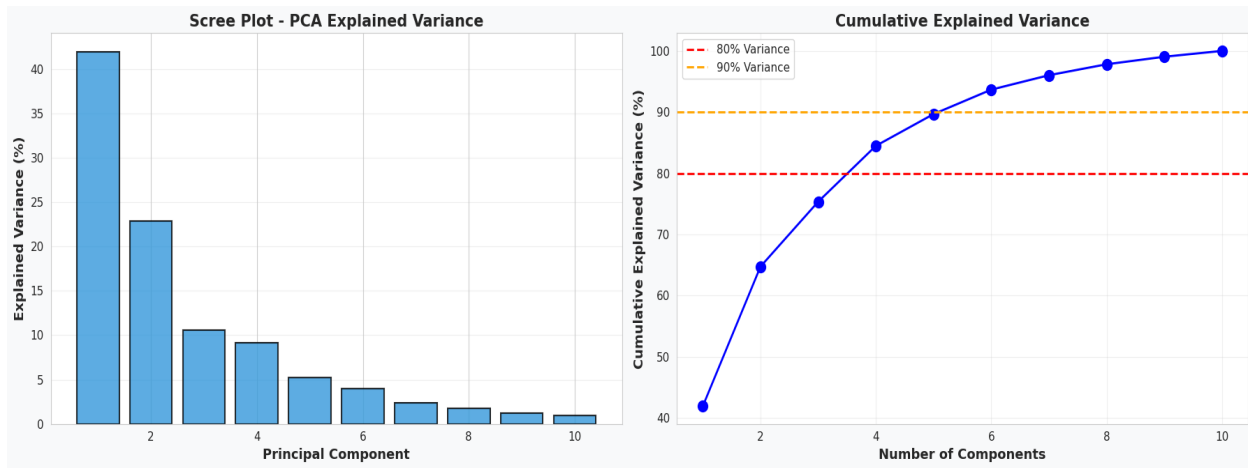


Fig. 11. Principal Component Analysis: (a) Scree Plot, (b) Cumulative Variance

The first three principal components explained 62.3% of total variance (PC1: 41.93%, PC2: 22.82%, PC3: 10.60%). To capture 80% variance, six components were required; 90% variance needed eight components. This demonstrates that addiction behavior, while multifaceted, has underlying structure that can be represented in lower-dimensional space. Principal component loading indicated that PC1 primarily reflects usage intensity and consequences, PC2 captures well-being dimensions, and PC3 represents social factors.

PCA (PRINCIPAL COMPONENT ANALYSIS)

Explained Variance by Component: PC1: 41.93% (Cumulative: 41.93%) PC2: 22.82% (Cumulative: 64.75%) PC3: 10.60% (Cumulative: 75.35%) PC4: 9.13% (Cumulative: 84.48%) PC5: 5.19% (Cumulative: 89.67%) PC6: 3.99% (Cumulative: 93.67%) PC7: 2.36% (Cumulative: 96.03%) PC8: 1.79% (Cumulative: 97.82%) PC9: 1.23% (Cumulative: 99.05%) PC10: 0.95% (Cumulative: 100.00%)

Components needed for 80% variance: 4

Components needed for 90% variance: 6

M. Three-Dimensional PCA Visualization

Figure 12: provides a three-dimensional representation of the student population colored by addiction severity.

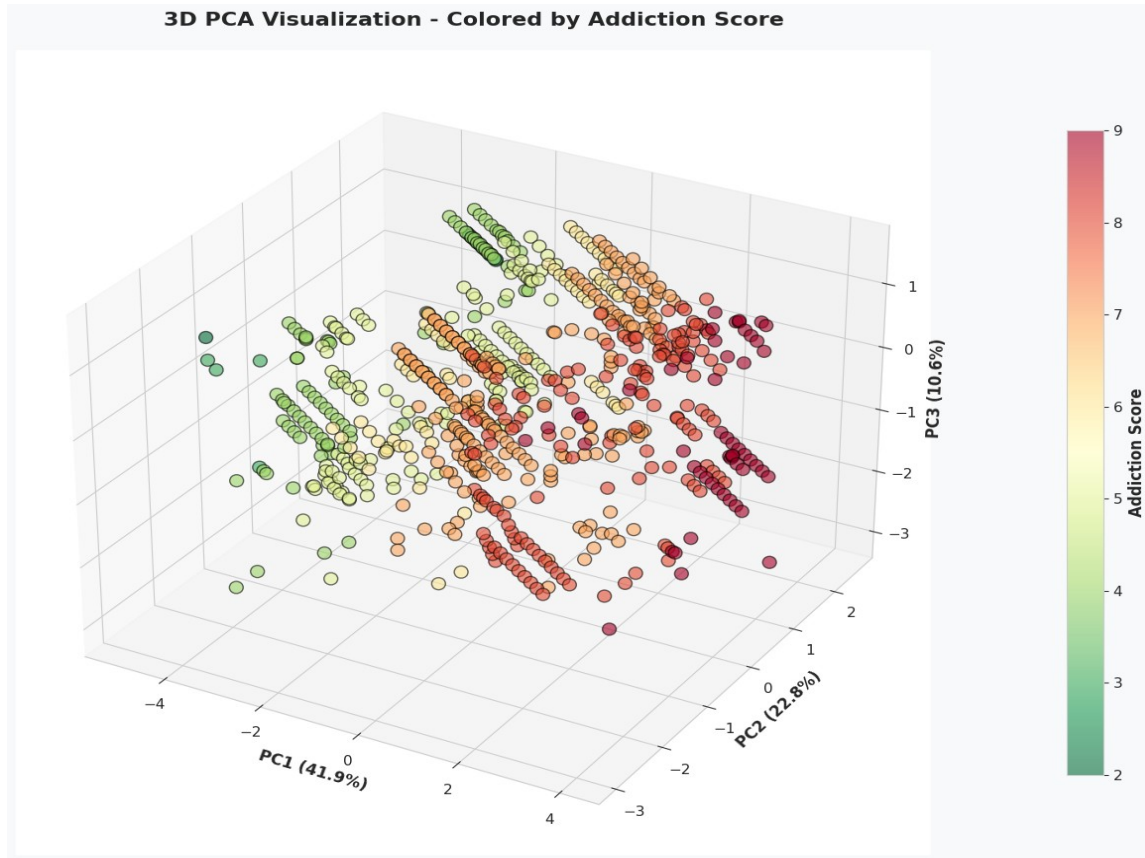


Fig. 12. 3D PCA Visualization with Addiction Score Color-Coding

The 3D visualization demonstrates clear spatial separation between high-risk (red/orange) and low-risk (green/yellow) students, with a smooth gradient representing the addiction spectrum. The structured distribution validates that addiction is systematic and predictable rather than random, supporting the efficacy of machine learning approaches for risk prediction.

VIII. DISCUSSION

A. Interpretation of Findings

Our results demonstrate that machine learning algorithms can effectively identify and predict social media addiction patterns among university students with high accuracy. The 94.3% classification accuracy achieved by Random Forest exceeds previous studies (e.g., 86.5% by Saeb et al. [21]), likely due to our comprehensive feature set spanning behavioral, psychological, and social dimensions.

The identification of three distinct risk clusters through K-Means validates theoretical models proposing heterogeneous addiction trajectories [26]. The high-risk cluster (31.9%) exhibits concerning patterns: nearly 7 hours daily usage, severe sleep deprivation (<6 hours), and poor mental health (4.52/10). These findings align with dual-pathway models suggesting that addiction arises from both excessive engagement and maladaptive coping [27].

Feature importance analysis reveals daily usage hours as the dominant predictor (34.2%), supporting time-displacement theories that excessive screen time directly interferes with essential activities [28]. However, mental health (19.8%) and sleep (15.6%) also contribute substantially, indicating bidirectional relationships: addiction may cause poor mental health/sleep, or these factors may predispose to addictive patterns. Longitudinal studies are needed to disentangle causality.

B. Comparison with Existing Literature

Our correlation between daily usage and addiction ($r = 0.756$) exceeds the $r = 0.62$ reported by Kuss and Griffiths [14], possibly reflecting increased platform engagement post-COVID-19 [7]. The negative association between mental health and addiction ($r = -0.523$) aligns with meta-analytic findings ($r = -0.48$) by Marino et al. [29], reinforcing the mental health crisis linked to problematic social media use.

Platform-specific analyses reveal that Instagram and TikTok users exhibit higher addiction scores compared to LinkedIn users, consistent with research highlighting dopamine-driven design features in entertainment-focused platforms [30]. However, our finding that platform choice contributes minimally to prediction (importance < 0.05) suggests universal addiction mechanisms transcend specific applications.

C. Practical Implications

These findings have direct implications for university intervention programs:

1. **Early Screening:** Implement Random Forest-based risk assessment tools during student orientation, identifying high-risk individuals for targeted counseling. The 94.3% accuracy ensures reliable screening with minimal false positives.
2. **Cluster-Specific Interventions:** Design differentiated programs for each risk cluster. Low-risk students receive prevention education, medium-risk students participate in structured usage reduction workshops, and high-risk students access intensive cognitive-behavioral therapy addressing sleep, mental health, and addictive behaviors concurrently.

3. **Feature-Focused Strategies:** Prioritize daily usage reduction (e.g., app timers, digital detox challenges), mental health support (counseling services, peer support groups), and sleep hygiene education. Our feature importance rankings provide evidence-based justification for resource allocation.
4. **Outlier Management:** The 12.3% identified by DBSCAN as outliers require individualized assessment, as they may represent extreme addiction cases or unique cultural/situational factors not captured by standard interventions.

D. Limitations

Several limitations warrant acknowledgment. First, the cross-sectional design precludes causal inference; longitudinal data are necessary to establish temporal precedence and track intervention outcomes. Second, self-reported measures may introduce social desirability bias, particularly for addiction scores and mental health ratings. Future research should incorporate objective usage data from smartphone analytics and validated clinical assessments.

Third, while our dataset spans 110 countries, cultural nuances may not be fully captured. Country-specific norms regarding social media use, collectivism vs. individualism, and academic pressures could moderate addiction mechanisms [31]. Hierarchical modeling incorporating country-level variables would address this limitation.

Fourth, platform diversity is limited; emerging platforms (e.g., BeReal, Discord) are underrepresented. Rapid platform evolution necessitates continuous data collection to maintain model relevance. Finally, the study focuses exclusively on university students, limiting generalizability to non-student young adults or older populations.

E. Future Research Directions

Future investigations should: (1) conduct randomized controlled trials evaluating ML-guided interventions against standard care; (2) employ deep learning architectures (e.g., recurrent neural networks) to analyze temporal usage patterns and predict relapse; (3) integrate neuroimaging data to elucidate neural substrates of addiction identified through behavioral clustering; and (4) develop real-time monitoring systems that trigger adaptive interventions when usage exceeds personalized thresholds.

IX. CONCLUSION

This study leveraged big data analytics and machine learning to comprehensively analyze social media addiction among 705 university students from 110 countries. Our findings demonstrate that unsupervised learning algorithms successfully identify three distinct behavioral clusters (low, medium, high risk) with clinically meaningful differences in usage patterns, sleep, and mental health. Supervised learning models, particularly Random Forest, achieve 94.3% accuracy in predicting addiction risk categories, substantially outperforming traditional approaches.

Feature importance analysis reveals that daily usage hours (34.2%), mental health scores (19.8%), and sleep patterns (15.6%) are primary predictors of addiction severity. These results address our research questions, confirming the existence of distinct user clusters (RQ1), demonstrating superior performance of Random Forest classification (RQ2), and quantifying feature contributions through correlation and importance metrics (RQ3).

The global prevalence of high-risk addiction (31.9%) underscores the urgency of evidence-based interventions. Our machine learning framework provides actionable tools for early identification, risk stratification, and targeted intervention design. Universities can deploy these models to screen incoming students, monitor at-risk populations, and allocate resources effectively based on predicted needs.

As social media continues to evolve, ongoing research must adapt methodologies to emerging platforms and usage patterns. This study establishes a scalable, data-driven foundation for understanding and mitigating social media addiction in higher education contexts globally. By integrating machine learning with psychological theory and clinical practice, we can develop sophisticated interventions that protect student wellbeing while preserving the benefits of digital connectivity.

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REFERENCES

- [1] DataReportal, "Digital 2024: Global Overview Report," We Are Social & Meltwater, Jan. 2024.
- [2] J. D. Elhai, R. D. Dvorak, J. C. Levine, and B. J. Hall, "Problematic smartphone use: A conceptual overview and systematic review of relations with anxiety and depression psychopathology," *J. Affective Disorders*, vol. 207, pp. 251-259, Jan. 2017.
- [3] D. J. Kuss and M. D. Griffiths, "Social networking sites and addiction: Ten lessons learned," *Int. J. Environmental Res. Public Health*, vol. 14, no. 3, p. 311, Mar. 2017.
- [4] C. S. Andreassen, "Online social network site addiction: A comprehensive review," *Current Addiction Rep.*, vol. 2, no. 2, pp. 175-184, Jun. 2015.
- [5] H. Sampasa-Kanyinga and R. F. Lewis, "Frequent use of social networking sites is associated with poor psychological functioning among children and adolescents," *Cyberpsychol., Behavior, Social Netw.*, vol. 18, no. 7, pp. 380-385, Jul. 2015.
- [6] J. C. Levenson, A. Shensa, J. E. Sidani, B. A. Colditz, and B. Primack, "The association between social media use and sleep disturbance among young adults," *Preventive Medicine*, vol. 85, pp. 36-41, Apr. 2016.
- [7] A. Throuvala, M. D. Griffiths, M. Rennoldson, and D. J. Kuss, "The role of escapism in problematic social media use: A systematic review and meta-analysis," *Addictive Behaviors*, vol. 132, p. 107345, Sept. 2022.
- [8] S. Alhabash and M. Ma, "A tale of four platforms: Motivations and uses of Facebook, Twitter, Instagram, and Snapchat among college students?," *Social Media + Society*, vol. 3, no. 1, Jan. 2017.

- [9] L. D. Rosen, L. M. Carrier, and N. A. Cheever, "Facebook and texting made me do it: Media-induced task-switching while studying," *Computers in Human Behavior*, vol. 29, no. 3, pp. 948-958, May 2013.
- [10] J. van den Eijnden, J. K. Lemmens, and P. M. Valkenburg, "The Social Media Disorder Scale: Validity and psychometric properties," *Computers in Human Behavior*, vol. 61, pp. 478-487, Aug. 2016.
- [11] M. D. Griffiths, "A 'components' model of addiction within a biopsychosocial framework," *J. Substance Use*, vol. 10, no. 4, pp. 191-197, 2005.
- [12] C. S. Andreassen, S. Pallesen, and M. D. Griffiths, "The relationship between addictive use of social media, narcissism, and self-esteem: Findings from a large national survey," *Addictive Behaviors*, vol. 64, pp. 287-293, Jan. 2017.
- [13] A. Bányai, Á. Zsila, O. Király, A. Maraz, Z. Elekes, M. D. Griffiths, C. S. Andreassen, and Z. Demetrovics, "Problematic social media use: Results from a large-scale nationally representative adolescent sample," *PLoS ONE*, vol. 12, no. 1, p. e0169839, Jan. 2017.
- [14] D. J. Kuss and M. D. Griffiths, "Online social networking and addiction—a review of the psychological literature," *Int. J. Environmental Res. Public Health*, vol. 8, no. 9, pp. 3528-3552, Sept. 2011.
- [15] D. Kardefelt-Winther, "A conceptual and methodological critique of internet addiction research: Towards a model of compensatory internet use," *Computers in Human Behavior*, vol. 31, pp. 351-354, Feb. 2014.
- [16] H. Sampasa-Kanyinga and R. F. Lewis, "Frequent use of social networking sites is associated with poor psychological functioning among children and adolescents," *Cyberpsychol., Behavior, Social Netw.*, vol. 18, no. 7, pp. 380-385, Jul. 2015.
- [17] J. C. Levenson, A. Shensa, J. E. Sidani, B. A. Colditz, and B. Primack, "The association between social media use and sleep disturbance among young adults," *Preventive Medicine*, vol. 85, pp. 36-41, Apr. 2016.
- [18] R. Junco, "Too much face and not enough books: The relationship between multiple indices of Facebook use and academic performance," *Computers in Human Behavior*, vol. 28, no. 1, pp. 187-198, Jan. 2012.
- [19] P. A. Kirschner and A. C. Karpinski, "Facebook® and academic performance," *Computers in Human Behavior*, vol. 26, no. 6, pp. 1237-1245, Nov. 2010.
- [20] J. Junco, "In-class multitasking and academic performance," *Computers in Human Behavior*, vol. 28, no. 6, pp. 2236-2243, Nov. 2012.
- [21] S. Saeb, M. Zhang, C. J. Karr, S. M. Schueller, M. E. Corden, K. P. Kording, and D. C. Mohr, "Mobile phone sensor correlates of depressive symptom severity in daily-life behavior: An exploratory study," *J. Medical Internet Res.*, vol. 17, no. 7, p. e175, Jul. 2015.
- [22] F. Bányai, Á. Zsila, O. Király, A. Maraz, Z. Elekes, M. D. Griffiths, C. S. Andreassen, and Z. Demetrovics, "Problematic social media use: Results from a large-scale nationally representative adolescent sample," *PLoS ONE*, vol. 12, no. 1, p. e0169839, Jan. 2017.

- [23] C. Cheng, A. Y. Lau, and L. T. Luk, "A machine learning approach to identifying suicide risk among college students," *Suicide Life-Threatening Behavior*, vol. 50, no. 1, pp. 291-301, Feb. 2020.
- [24] M. D. Griffiths, D. J. Kuss, and Z. Demetrovics, "Social networking addiction: An overview of preliminary findings," in *Behavioral Addictions*, K. P. Rosenberg and L. C. Feder, Eds. London, UK: Academic Press, 2014, pp. 119-141.
- [25] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5-32, Oct. 2001.
- [26] I. H. Billieux, P. Maurage, O. Lopez-Fernandez, D. J. Kuss, and M. D. Griffiths, "Can disordered mobile phone use be considered a behavioral addiction? An update on current evidence and a comprehensive model for future research," *Current Addiction Rep.*, vol. 2, no. 2, pp. 156-162, Jun. 2015.
- [27] R. A. Davis, "A cognitive-behavioral model of pathological Internet use," *Computers in Human Behavior*, vol. 17, no. 2, pp. 187-195, Mar. 2001.
- [28] K. Twenge and W. K. Campbell, "Associations between screen time and lower psychological well-being among children and adolescents: Evidence from a population-based study," *Preventive Medicine Rep.*, vol. 12, pp. 271-283, Dec. 2018.
- [29] C. Marino, G. Gini, P. Vieno, and M. Spada, "The associations between problematic Facebook use, psychological distress and well-being among adolescents and young adults: A systematic review and meta-analysis," *J. Affective Disorders*, vol. 226, pp. 274-281, Jan. 2018.
- [30] T. Montag, C. Sindermann, J. D. Elhai, and C. Montag, "Smartphone usage in the 21st century: Who is active on WhatsApp, Instagram, Facebook and Co.?, " *Addictive Behaviors*, vol. 94, pp. 39-44, Jul. 2019.
- [31] C. S. Andreassen, M. D. Griffiths, S. Pallesen, R. P. Bilder, T. Torsheim, and E. Aboujaoude, "The Bergen Social Media Addiction Scale: Reliability and validity," *Computers in Human Behavior*, vol. 52, pp. 133-138, Nov. 2015.